Questions

Q1.

A motorcycle is travelling at a velocity of 6.2 m/s.

The motorcycle accelerates at 2.5 m/s^2 until its velocity is 10 m/s.

(i) Calculate the time taken for this acceleration.

Use the equation

$$\text{time taken} = \frac{\text{change in velocity}}{\text{acceleration}}$$

(2)

time taken = ... s

(ii) The motor cycle now decelerates (slows down) from 10 m/s to a stop.

The deceleration is at a constant rate of 4.4 m/s^2.
Calculate the distance the motorcycle travels as it slows down to a stop.
Use the equation

$$v^2 - u^2 = 2 \times a \times x$$

(2)

distance = ... m

(Total for question = 4 marks)

Q2.

Figure 10 shows a small steel ball held at a height, *h*, above the ground.

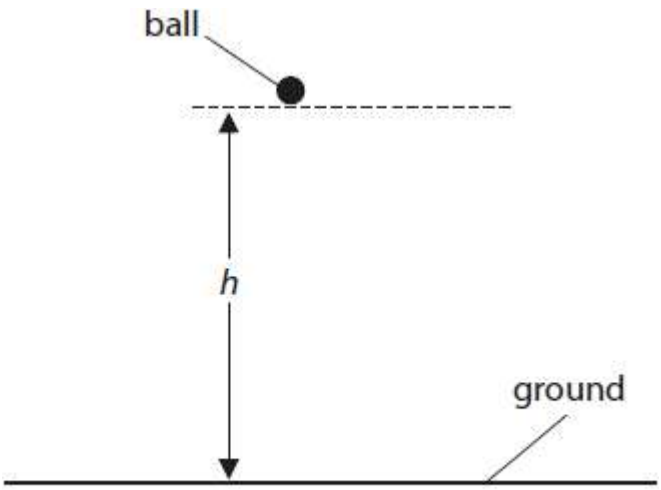

Figure 10

The ball is released and allowed to fall to the ground.

The height *h* is 1.4 m.

Calculate the time, *t*, for the ball to reach the ground.

Use the equation

$$t^2 = \frac{2h}{g}$$

g = 10 m/s^2

(2)

t = ... s

(Total for question = 2 marks)

Q3.

An atom of mass 6.6 × 10^{-26} kg is moving with a velocity of 480 m / s.

Calculate the momentum of the atom.

(3)

momentum = ... kg m/s

(Total for question = 3 marks)

Q4.

Figure 12 shows a ball held in a clamp at **R**, above the ground.

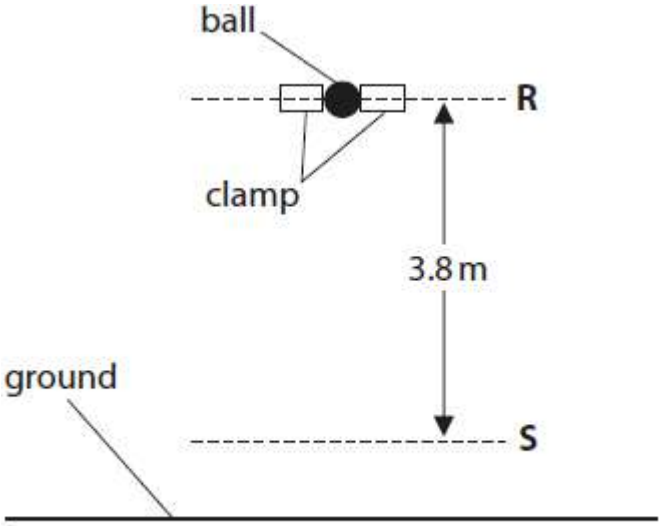

Figure 12

The ball is released from the clamp and falls.

S is 3.8 m below **R**.

At **S** the momentum of the ball is 0.40 kg m/s.

Calculate the mass of the ball.

Acceleration due to gravity, g, = 10 m / s^2

(4)

mass of the ball ..... kg

(Total for question = 4 marks)

Q5.

Figure 3 is a velocity-time graph for the motion of a van on a long, straight road.

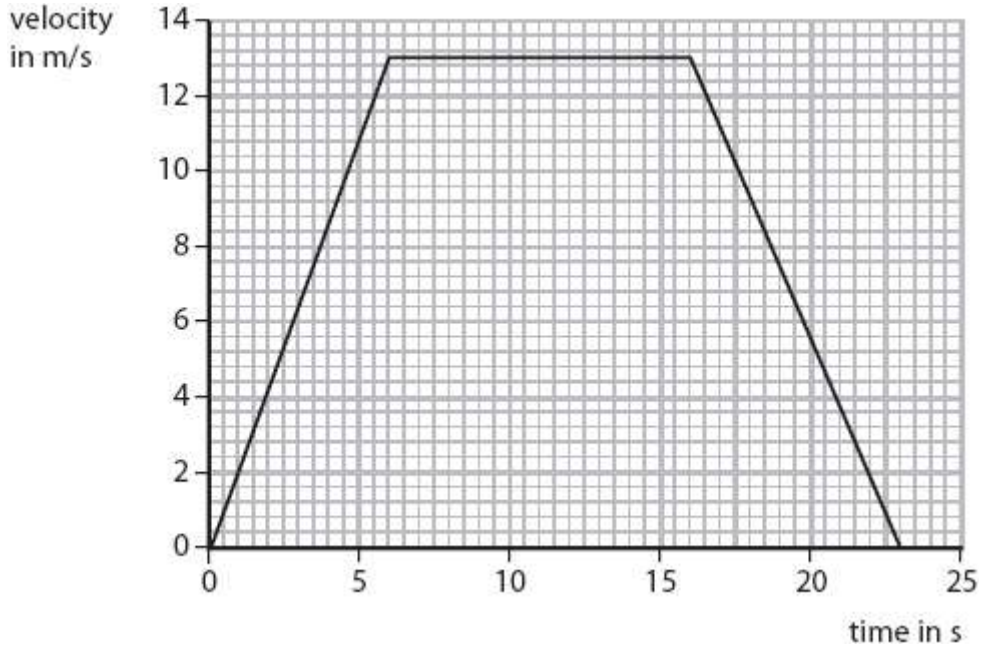

Figure 3

(i) The van accelerates at 2.2 m/s^2.

The force causing this acceleration is 5.5 kN.
Calculate the mass of the van.

(3)

mass = ... kg

(ii) Calculate the distance travelled by the van between 16.0 s and 23.0 s.

Give your answer correct to 2 significant figures.

(4)

distance travelled = ... m

(Total for question = 7 marks)

Q6.

Figure 11a shows a box falling towards a hard floor.

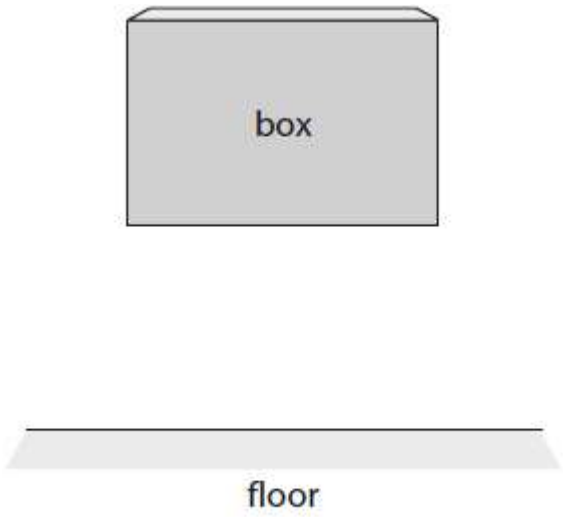

Figure 11a

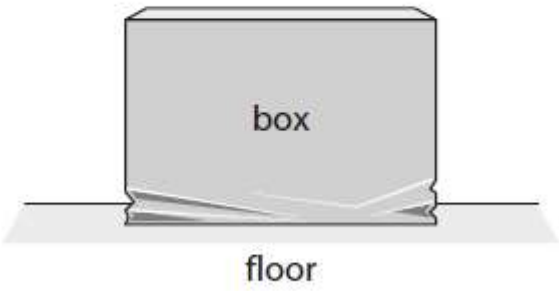

Figure 11b

The box hits the floor and crumples a little before it comes to rest as shown in Figure 11b.

The momentum of the box just before it hits the floor is 8.7 kg m/s.

The box comes to rest 0.35 s after it first hits the floor.

(i) Calculate the magnitude of the force exerted by the floor on the box.
Use an equation selected from the list of equations at the end of this paper.

(2)

force exerted by the floor on the box = ... N

(ii) State the magnitude and direction of the force exerted by the box on the floor.

(2)

magnitude .....

direction .....

(Total for question = 4 marks)

Q7.

The **inertial** mass of an object is a measure of how difficult it is to change the velocity of the object.

A force of 450 N acts on a car to give the car an acceleration of 0.35 m/s^2.

Calculate the **inertial** mass of the car.

(2)

inertial mass of car ..... kg

(Total for question = 2 marks)

Q8.

Figure 18 shows a ball before and after it collides with a wall.

The arrows show the direction of movement of the ball.

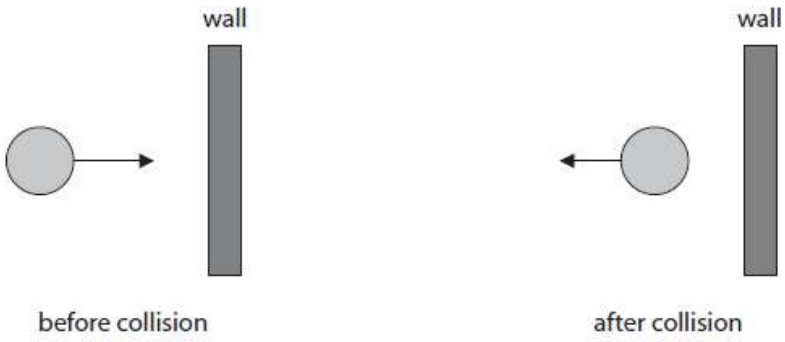

Figure 18

Before the collision, the momentum of the ball is 0.80 kg m / s.

After the collision, the momentum of the ball is 0.60 kg m / s in the opposite direction.

The ball is in contact with the wall for a time of 70 ms during the collision.

Calculate the force exerted on the ball by the wall.

(3)

Use an equation selected from the list of equations at the end of the paper.

force = ... N

(Total for question = 3 marks)

Q9.

Figure 8 is a velocity/time graph for a lift moving upwards in a tall building.

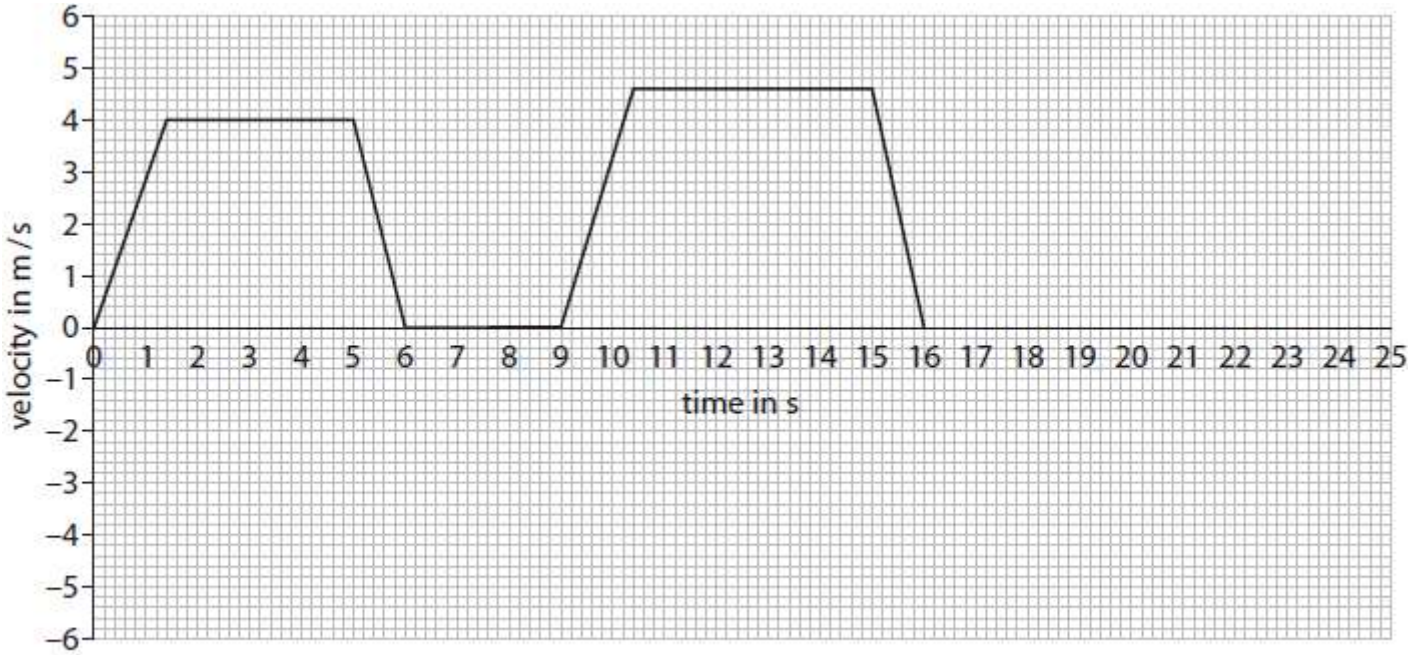

Figure 8

Use the graph in Figure 8 to determine the distance that the lift travelled during the first 6.0 s.

(3)

distance = ... m

(Total for question = 3 marks)

Q10.

Figure 12 shows a skier on a slope.

The skier travels down the slope with a constant acceleration.

The speed of the skier is measured at points P and Q.

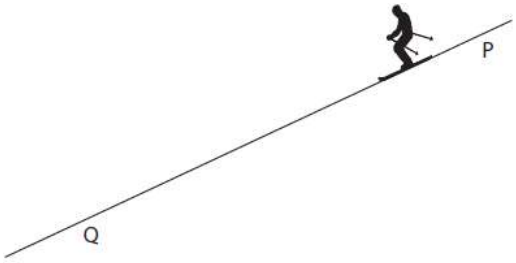

Figure 12

The table in Figure 13 gives some data about the skier making one downhill run.

acceleration	3.0 m/s^2
speed at P	7.6 m/s
speed at Q	24 m/s

Figure 13

(i) Calculate the distance from P to Q.

Use an equation selected from the list of equations at the end of this paper.

(3)

distance from P to Q = ... m

(ii) Calculate the time taken for the skier to travel from P to Q.

(3)

time from P to Q = ... s

(Total for question = 6 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (t =) $\frac{10 - 6.2}{2.5}$	$\frac{3.8}{2.5}$ allow $\frac{6.2 - 10}{2.5}$ or $\frac{-3.8}{2.5}$	2 AO2.1
	evaluation (1) (t =) 1.5 (s)	1.52 (s) allow –1.5(2) (s) award full marks for correct answer without working	

Question number	Answer	Additional guidance	Mark
(ii)	substitution OR rearrangement (1) $(-)10^2 = 2 \times (-)\ 4.4 \times x$	$(x =)\ \frac{v^2 - u^2}{2 \times a}$ $(x =)\ \frac{(-)10^2}{2 \times (-)\ 4.4}$	2 AO2.1
	evaluation (1) (x =) 11 (m)	allow values that round to 11	

		(m) e.g. 11.36 (m) ignore negative sign in answer line accept 1.1(36) for one mark award full marks for correct answer without working	

Q2.

Question number	Answer	Additional guidance	Mark
CS5	substitution (1) $(t^2 =)\ \frac{2 \times 1.4}{10}$ evaluation (1) $(t =)$ 0.53 (s)	0.28 allow numbers that round to 0.53 e.g. 0.52915 award full marks for correct answer without working.	**(2)** **AO2**

Q3.

	Answer	Additional guidance	Mark
	selection (1) $p = m \times v$ substitution (1) $6.6 (\times 10^{-26}) \times 480$ evaluation (1) 3.2×10^{-23} (kg m/s)	allow mom(entum) = mass x velocity allow numbers that round to 3.2×10^{-23} e.g. 3.168×10^{-23} award full marks for the correct answer without working $6.6 (\times 10^{-26}) \times 480$ seen scores MP1 and MP2, 2 marks 3.2 to any other power of ten scores MP1 and MP2, 2 marks	**(3)** **AO2**

Q4.

Question number	Answer	Additional guidance	Mark
	Two stage calculation		**(4) AO2**
	substitution$_1$ (1)	use of $v^2 - u^2 = 2ax$ OR $\frac{1}{2} mv^2 = mgh$	
	$(v^2 - 0 =)\ 2 \times 10 \times 3.8$	76	
	evaluation of v (1)		
	$(v =)$ 8.7 (m/s)	allow numbers that round to 8.7 e.g. 8.718	
	substitution$_2$ (1)		
	$0.40 = m \times 8.7$	use of $p = mv$	
	rearrangement and evaluation (1)		
	$(m =)$ 0.046 (kg)	allow numbers that round to 0.046 e.g. 0.04598 award full marks for correct answer without working.	

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	rearrangement (1) (m=) $\frac{F}{a}$ substitution (1) (m=) $\frac{5500}{2.2}$ evaluation (1) 2500(kg)	Rearrangement and substitution may be in either order (m=) $\frac{5.5}{2.2}$ ignore unit conversion until evaluation POT (power of ten) error, e.g. 2.5 (kg) award 2 marks award full marks for correct numerical answer without working If no other marks scored, award one mark for a clear conversion of units	(3)

Question Number	Answer	Additional guidance	Mark
(ii)	correctly identifies data points from the graph to calculate areas (1) use of distance = area under graph (1) substitution (1) (distance) = $\frac{(23-16)\times 13}{2}$ evaluation rounded to 2SF (1) 46 (m)	both 7 and 13 seen (distance) = $\frac{7\times 13}{2}$ (= 45.5) accept 45 (m) award full marks for correct numerical answer without working 91 scores 2 marks (incorrect area calculation using distance x time)	(4)

Q6.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (force =) $\frac{8.7}{0.35}$ evaluation (1) 25 (N)	 use of force = $\frac{\text{change in momentum}}{\text{time}}$ allow numbers that round to 25 e.g 24 .8571 award full marks for correct answer without working.	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	(magnitude) 25 (N) (1) (direction) down(wards)/ towards floor (1)	ecf from 7bi allow arrow drawn pointing down "south"	**(2)** **AO3**

Q7.

Question	Answer	Additional guidance	Mark
	identification of equation and substitution (1) 450 = m x 0.35 rearrangement and evaluation (1) (m=) $\frac{450}{0.35}$ 1300 (kg)	 accept values which round to 1300 (kg) e.g. 1286 (kg) award full marks for the correct answer without working	**2** **AO2.1**

Q8.

	Answer	Additional guidance	Mark
	attempt to find momentum change (Δp) (1) (Δp) =± 1.4 substitution in to $F = \frac{\Delta p}{t}$ (1) $\frac{\pm 1.4}{70 \times 10^{-3}}$ evaluation (1) (±) 20 (N)	± 0.8 ± 0.6 allow (Δp) =± 0.2 allow $\frac{\pm 0.2}{70 \times 10^{-3}}$ answers which round to (±) 20 to any other power of 10 score 2 marks answers which round to (±) 2.9 scores 2 marks answers which round to (±) 2.9 to any other power of 10 score 1 mark award full marks for the correct answer without working	**(3)** **AO2**

Q9.

	Answer	Additional guidance	Mark
	(distance =) area (under graph) (1) substitution (1) ½(1.4 x 4) + (3.6 x 4) + ½(1x 4) evaluation (1) 19 (m)	may be seen on graph 2.8 + 14.4+ 2.0 ½ x [3.6+6] x 4 allow values that round to 19 (m) (e.g. 19.2..) award full marks for the correct answer without working if no other marks scored allow (4 x 6 =)24 (m) for 1 mark	**(3)** **AO3**

Q10.

Question Number	Answer	Additional guidance	Mark
(i)	substitution in $v^2 - u^2 = 2ax$ (1) $24^2 - 7.6^2 = 2 \times 3 \times x$ rearrangement (1) $(x =) \frac{24^2 - 7.6^2}{6}$ evaluation (1) 86 (m)	accept rearrangement and substitution in either order allow numbers that round to 86 (m) award full marks for the correct answer without working	**(3)**

Question Number	Answer	Additional guidance	Mark
(ii)	recall and substitution (1) $(a = \frac{v - u}{t})$ $3.0 = \frac{24 - 7.6}{t}$ rearrangement (1) $t = \frac{v-u}{a}$ OR $(t =) \frac{24 - 7.6}{3.0}$ evaluation (1) 5.5 (s)	Allow alternative method: average speed = distance / time i.e. 15.8 = 86(.37) / time (t =) 86(.37) / 15.8 allow numbers that round to 5.5 (s) OR numbers that round to 5.4 if using alternative method and distance = 86 award full marks for the correct answer without working no marks for t = d / (v-u) = 86(.37) / (24-7.6) giving 5.3 s as an answer	**(3)**